



TRIBHUVAN UNIVERSITY
FACULTY OF HUMANITIES AND SOCIAL SCIENCES

ONLINE CHAT APPLICATION
(MeroChat)

A PROJECT REPORT

Submitted to:

Department of Computer Application
Mechi Multiple Campus
Bhadrapur, Jhapa

***In partial fulfillment of the requirements for the Bachelors in Computer
Application***

Submitted By:

Manoj Sitoula

BCA 4th Semester

University SN: 6-2-0002-0062-2018

Symbol No.: 18020017

.....
Under the Supervision of

.....



TRIBHUVAN UNIVERSITY
FACULTY OF HUMANITIES AND SOCIAL SCIENCE
Mechi Multiple Campus
Bhadrapur, Jhapa

Bachelor in Computer Applications (BCA)

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by **Manoj Sitoula** entitled “**MeroChat**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

.....

Project Supervisor
BCA Department
Mechi Multiple Campus



Tribhuvan University

Faculty of Humanities and Social Sciences

Mechi Multiple Campus

Bhadrapur, Jhapa

Bachelor in Computer Applications (BCA)

LETTER OF APPROVAL

This is certify that this project prepared by **Manoj Sitoula** entitled “**MeroChat**” in the partial fulfillment of requirement for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

.....

Supervisor
BCA Department
Mechi Multiple Campus

.....

External Examiner

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Krishna Prasad Acharya
Director
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.....

Internal Examiner

ABSTRACT

In this project, we intend to develop a web-based chat application, which will allow users to communicate with each other over internet. It overcomes old methods of communication between users by making simple interface to communicate with each other.

Previously, people living apart from each other communicate with each other writing emails as this system wasn't real-time, we had to wait for reply of email from another user for days. This would be very annoying. But, by using MeroChat people can communicate with each other in real time from any point of the world as if they are in front of each other. Here, users can have one to one conversation with each other, can transfer files and pictures. Users can click and send photos. Users can send and delete messages, messages are deleted permanently on both sides. Here one user can search for another user using their name. Users can update their profiles; they can view each other profiles. It also shows the status of user's whether they are online or offline and either message status are seen or unseen.

ACKNOWLEDGEMENT

We would like to express our special thanks of gratitude to our supervisor **Mr.....** who gave us the golden opportunity to do this wonderful project on the topic of **Online Chat Application (MeroChat)**, which also helped us in doing a lot of research and we came to know about so many new tools and technologies.

We would like to express our special thanks of gratitude to our Campus Chief **Mr. Jivan Pokharel** who gave us permission for doing this Project.

I would like to express my special gratitude and thanks to our BCA Program Coordinator **Mr. Krishna Prasad Acharya** for his support and help for our personnel development and mainly for the completion of this Project.

I am highly indebted to Mechi Multiple Campus for their guidance and constant supervision as well as for providing necessary information regarding the Project and support in the completion.

We would also like to express my gratitude towards library and member of Mechi Multiple Campus for their kind co-operation and encouragement which help me in completion of this Project.

We would also like to thank our parents and friends who helped us a lot in finalizing this project within the limited time frame.

In the end, we would also like to thank Tribhuvan University for giving us this opportunity via the course of Computer Application to help us understand the project ethics at this early stage and helped us to evaluate my knowledge and expand it a little more.

Yours sincerely,
Manoj Sitoula

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LIST OF ABBREVIATIONS

1. DFD: Data Flow Diagram.
2. ERD: Entity Relationship Diagram.
3. MVC: Model View Controller.
4. CSS: Cascade style sheet.
5. HTML: Hypertext markup language.
6. AJAX: Asynchronous JavaScript and XML
7. XML: Extensive Markup Language
8. PHP: Hypertext Preprocessor.
9. MySQL: Structured Query Language.

CHAPTER 1

INTRODUCTION

1.1 Introduction

Humans are social animal who cannot live without communicating with each other. Everyone communicates with each other using various means of communication like Telephone, Television, and Mobile. But in today's world there are many new problems being created with the advancement of technology everything can be hacked. There are many applications from where people get unauthorized access to the system and leaks people's personal data. In today's world there is no proper privacy in using internet. No one knows when who will get hacked. Not only hackers but some company too steals personal data from their users. So, in order to maintain security for personal data, chats and others files, we got ourselves motivated to develop a system where users can use internet and can chat and share files and photos with each other without being scared of being hacked by any other people and getting files and data lost. So, this project is to create a chat application where users can have conversation with each other and share files and photos securely.

1.2 Problem Statement

In today's world there are many applications developed but there is lack of proper security to the data and information that are being shared through the means of internet. Many application too share users data and information to third party companies to run ads in their application.

So in order to minimize all this risk and make our application more suitable for using to the simple user we have developed an chat system where user can have private one to one chat using secure end to end encryption and ads free communication. Communication is also made fast and simple secure.

1.3 Objectives

- To create simple platform for communication between users.
- To allow users, send and receive files and photos.
- To allow users to have private one to one chat.

1.4 Scope and Limitation

1.4.1 Scope

Under this time span divided for this project I aim to complete a basic and functional prototype for what I target to build in future. In this development, this project will be covering following areas:

- Modules for registration and login of user.
- Modules for viewing and updating profile.
- Platform for typing and sending message.
- Platform for clicking and sending photo.
- Feature of showing online or offline status.
- Platform for searching friends.

1.4.2 Limitation

- Needs active internet connectivity
- Needs web enabled device.
- It does not support audio and video calls.
- It does not support voice message.

1.5 Report Organization

The material presented in the project is organized into five chapters:

Introduction

This chapter deals with the introduction of the system with its objectives and limitation along with the reason why the system is made.

Background Study and Literature Review

This chapter describes the fundamental theories and concepts as well as information about existing system, journals and references.

System Analysis and Design

This chapter focuses on the different requirement of the system, which describes about the functional, non-functional, feasibility analysis, Entity Relational diagram, Data Flow Diagram, design of the system with system architecture, database schema, and interface design.

Implementation and Testing

This chapter will provide an account on implementation and testing, tools used for preparation of the project. Test cases for unit testing as well as integration testing are done. Implementation details of modules will be trace.

Conclusion and Future Recommendation

This chapter will be the brief summaries of outcome of the project, conclusion, reviews as well as future recommendations, improvements that can be done on upcoming days and feedback systems, stability of the project.

CHAPTER 2

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Communication is a means for people to exchange their emotions, ideas, and feelings. It all started at the beginning of human creation. Distant communication began as early as the 18th century with the introduction of television, telegraph, and then telephone. Telephone communication stands out as the fastest growing technology, from fixed line to mobile wireless, from a voice call to data transfer. The emergence of computer networks and telecommunication technologies bears the same objective that is to allow people to communicate. All this while many efforts have been drawn towards consolidating the device into one and therefore indiscriminate the services. Chatting is the method of using technology to bring people and ideas together despite geographical barriers. The technology has been available for years, but the acceptance was quite recent. Our project is an example of a chat application. It is made using Ajax and MySQL. To start chatting user just have to open application where they can have conversation privately one-to-one

2.2 Literature Review

Since there are many chatting application available in the market, some of most existing communicating applications are Facebook's Messenger and Email.

Messenger is one of the most widely used instant messaging service in the world now. almost every country allows this services in their territory.it was founded as Facebook Chat in 2009 by Meta Platforms. It has many advantages as well as dis advantages. Some of its advantages are server are easily available ,fast and reliable.[1]

Whats app is another messaging app available in the today's world. It was founded in 2009 by Brain Action and Jan Koum. It is also most widely used messaging services available in south Asia. It provides end to end encryption messaging and video calling service.[2][3].

CHAPTER 3:

SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

This system is designed with the series of processes starting with requirement analysis, design, implementation, testing and maintenance. During requirement analysis, all the functional and nonfunctional requirement are analyzed and system is developed according to the requirement then designing of the system is carried out. After the design process, coding and development part is started then after integrating the system there is testing of the system. If the testing is positive then system is implemented otherwise some maintenance is done and system come in operation.

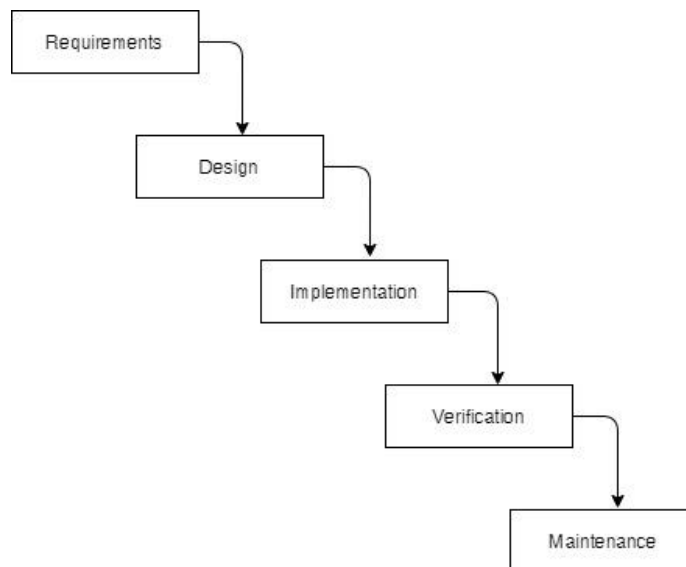


Figure 3-1: Waterfall Methodology for MeroChat

3.1.1 Requirement Analysis

The requirements are to be collected before starting project's development life cycle. To design and develop system, functional as well as non-functional requirement of the system has been studied.

i. Functional Requirement

- User must be able to register and login.

- User must be able to send messages, files and photos.
- User must be able to see online or offline status.
- User must be able to see if their message is seen or not seen.
- User must be able to access camera and click photo.
- User must be able to see and update their profiles.

USECASE DIAGRAM

In MeroChat, there is only one actor that is user where user can register , login, send messages, files, update their profile, search friends, change password, click photos and logout from their account.

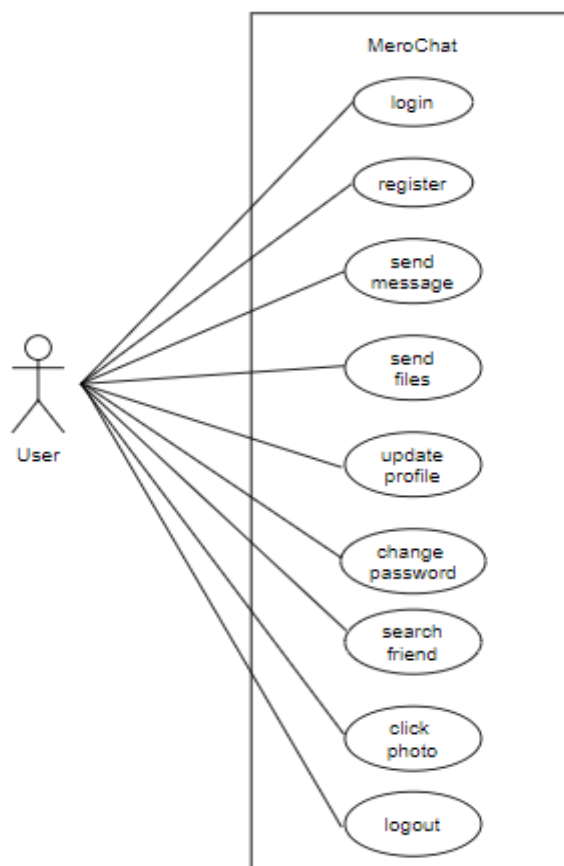


Figure 3-2: Use Case Diagram of MeroChat

ii. Non-functional requirement

Different non-functional requirement have been studied and identified and are listed as below:

- User Interface.
- Availability.
- Usability.
- Reliability.
- Responsiveness.

3.1.2 Feasibility analysis

- **Technical feasibility study**

The system is technically feasible as the requirement for the development of the system is easily accessible. The necessary hardware and software required for the development and implementation of the system is available. The basic programming language which is suitable for project is available and the libraries required for project is capable of achieving the result that we are aiming for. All the existing resources can be used for the development and maintenance system.

- **Economic feasibility study**

The system is economically feasible and cost effective. As all the tools and resources required are either open sources or free. After the completion of the system organization didn't need to deploy any new hardware and software as the required software and hardware. The existing resource of the system can be used.

- **Operational feasibility study**

The system is easy to operate with the basic knowledge of computer and internet and well trained manpower is not necessary. User can also easily access the system as it is user friendly in many aspects with good User Interface (UI). This system include all the requirements used for Chat system and this system is completely operational and can be successfully implemented.

- **Schedule feasibility study**

The deadline of project is set 12th September 2021. The time frame provided is tough and is immovable yet the project requirement and technology being used are compatible and the project can be done with proper planning. The project team

planned to accomplish the project within the time frame following the plan which is listed in the table below and shown in the Gantt chart. The project team concluded that following the schedule is planned, the project is Schedule feasible.

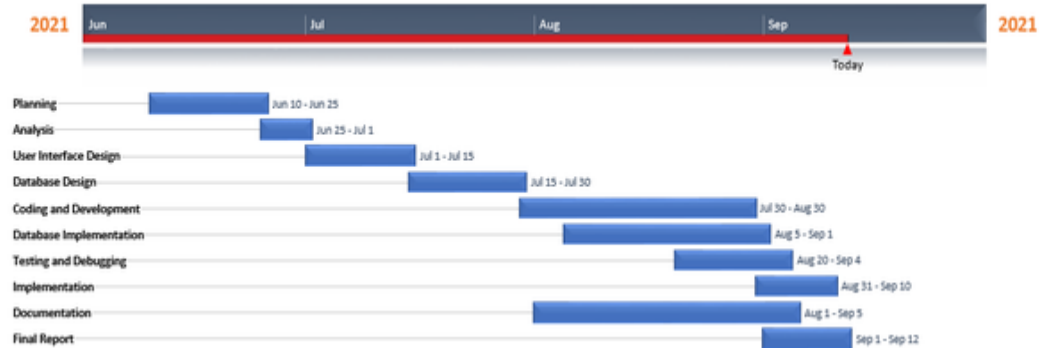


Figure 3-3: Gantt Chart of MeroChat

Table 3.1: Project Timeline of MeroChat

Title	T/M ↑	Start	End	🕒
Planning	T	6/10/2021	6/25/2021	12 days
Analysis	T	6/25/2021	7/1/2021	5 days
User Interface Design	T	7/1/2021	7/15/2021	11 days
Database Design	T	7/15/2021	7/30/2021	12 days
Coding and Development	T	7/30/2021	8/30/2021	22 days
Database Implementation	T	8/5/2021	9/1/2021	20 days
Testing and Debugging	T	8/20/2021	9/4/2021	11 days
Implementation	T	8/31/2021	9/10/2021	9 days
Documentation	T	8/1/2021	9/5/2021	25 days
Final Report	T	9/1/2021	9/12/2021	8 days

3.1.3 Data Modeling (Entity Relationship)

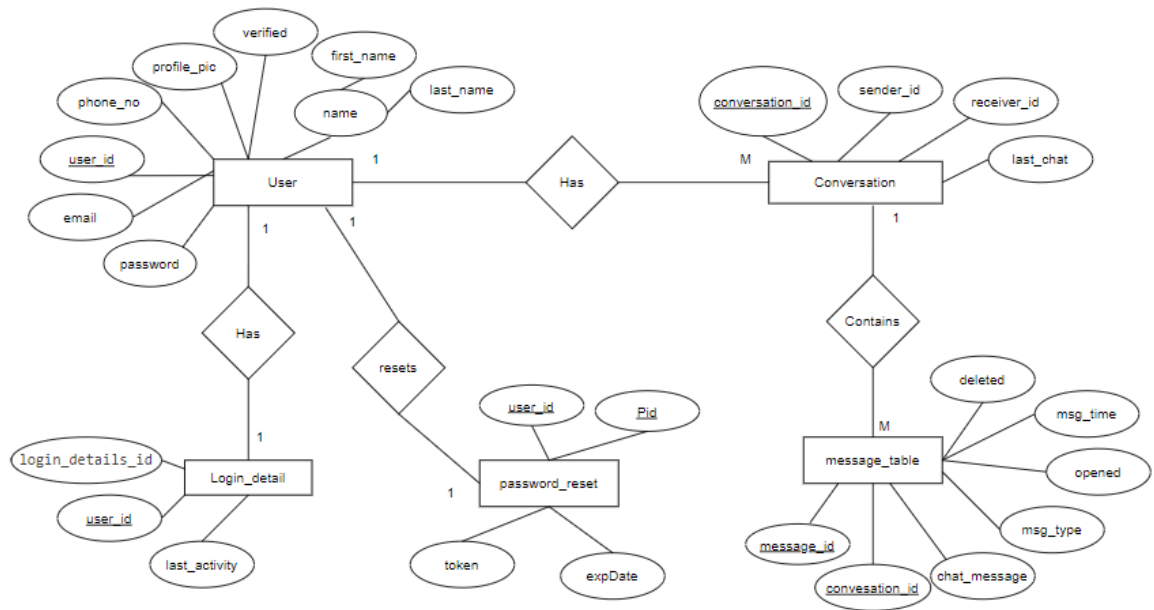


Figure 3-4: ER-D of MeroChat

ER-D Schema

- 1) User(first_name, last_name, verified, profile_pic, phone_no, user_id, email, password)
- 2) Conversation(conversation_id, sender_id, receiver_id, last_chat)
- 3) Message_table(message_id, conversation_id, chat_message, msg_type, opened, msg_time, deleted)
- 4) Password_reset(user_id, pid, token, expDate)
- 5) Login_detail(login_details_id, user_id, last_activity)

3.1.4 Process Modeling (DFD)

Data Flow Diagram of MeroChat consists of two levels of DFD context diagram and level one DFD. Both these levels are used for making data flow diagram of MeroChat.

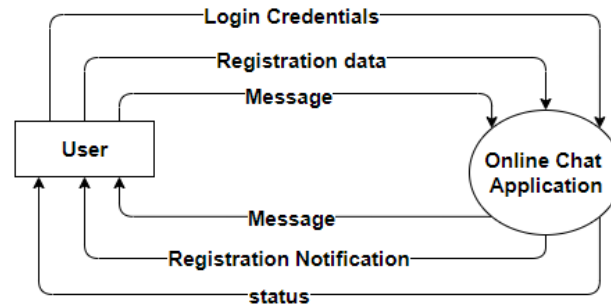


Figure 3-5: Context Diagram of MeroChat

In context diagram, Login credentials, registration data and message are the input of Online Chat Application (MeroChat) where users request for registration, login and sending message using chat mechanism. Message, registration notification and status are the output of Online Chat Application (MeroChat). Where users get registration notification after successful registration and receive message sent from friends with their status on the application.

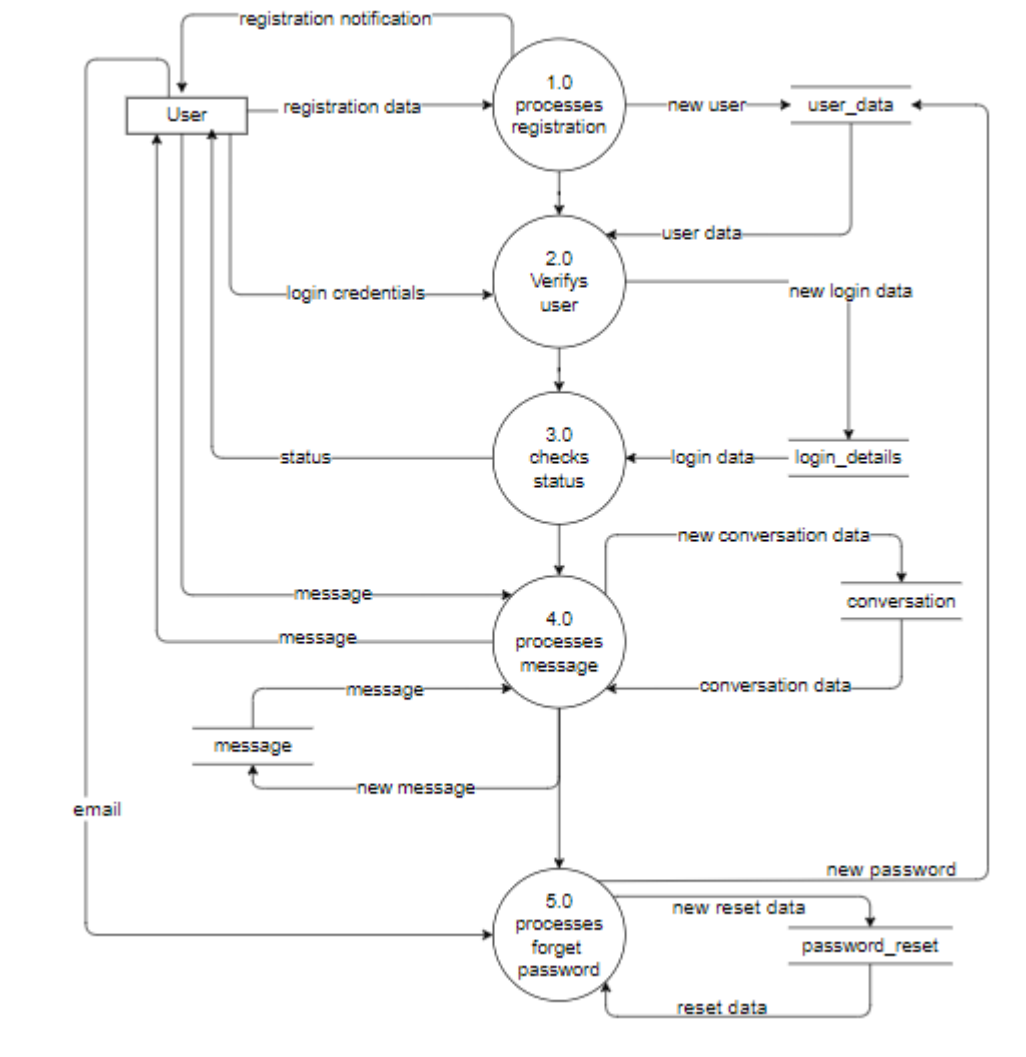


Figure 3-6:Level 1 Data Flow Diagram of MeroChat

In level 1 DFD, there are five processes where process 1 is responsible for registration of new user in the application, process 2 is responsible for checking login credentials to give access to the existing user to use the system, process 3 is responsible for checking status of user, process 4 is responsible for processing incoming and outgoing messages in the application, and process 5 is responsible for managing forget password requests.

3.2 System design

System design is the process of designing the elements of a system such as the architecture, modules and components, the different interfaces of those components

and the data that goes through that system. MeroChat is designed using various schema design, CASE tools and is described using following designs.

3.2.1 Architectural design

Architectural design represents the structure of data and program components that are required to build a computer based system, it is not an operational but it is a representation of system. For this system, three tier architecture is used which includes user interface/client tier, web server/web tier and database/data tier. In architectural design, basic structure of the system is shown.

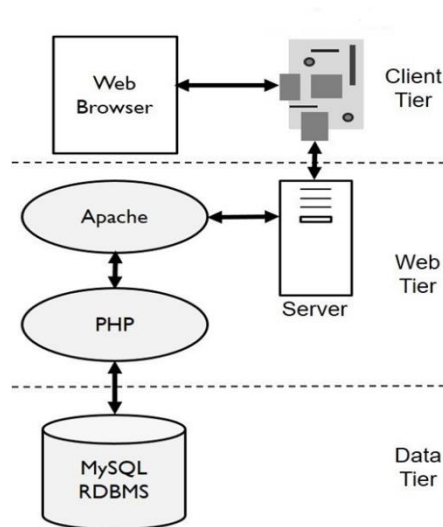


Figure 3-7:Architecture diagram of MeroChat

3.2.2 Database Schema Design

Database schema design organizes the data into separate entities, determines how to create relationships between organized entities, and how to apply the constraints on the data. Designers create database schemas to give other database users, such as programmers and analysts, a logical understanding of the data. The diagram shows identified tables, their columns and relationships among tables. According to our figure five tables are identified with their primary keys:

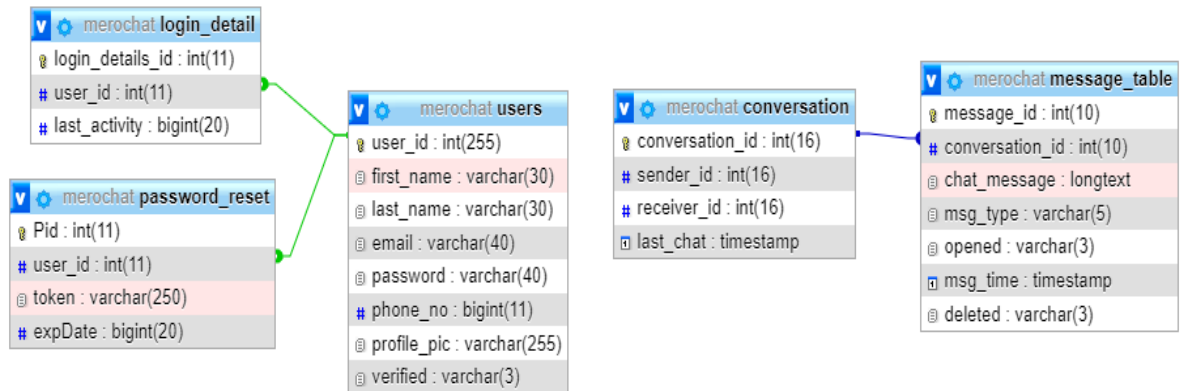


Figure 3-8: Schema Diagram of MeroChat

3.2.3 Interface Design

First of all, when user open website it will redirect user towards login page. If user is already registered and verified then they can simply login using their credentials. If user is not registered yet, user can click on Sign up link to redirect themselves in Sign up page. After registering themselves, user have to verify their email by clicking link that is sent to their email.

After verifying email and logging into the system, user will land on homepage where list of recent chats and friend list is shown. Users can click on any users and can have conversation. If they have no conversation yet then they can navigate to friends tab and can search users with their name. User can view their profiles, update them, change password and logout too from same page navigating to the top right corner of interface.

After clicking name of user, user will be redirected to chat interface where users can send and receive messages. User can send files and photos, can click picture as well and can send a message to friend. User can view their friends profile, can delete chat, and can view history of shared files. Users can download files and photos shared by clicking on download icon.

CHAPTER 4:

IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1. Tools Used

Following are the tools and framework used for the accomplishment of this project:

Front End Tools

- **HTML 5 (Hyper Text Mark-up Language):**

HTML is the set of markup symbols or codes inserted into a file intended for display on the Internet. The markup tells web browsers how to display a web page's words and images.

- **CSS 3 (Cascading Style Sheet):**

CSS is a style sheet language used for describing the presentation of a document written in a markup language such as HTML. CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript.

- **JavaScript :**

It is high-level, often just-in-time compiled, and multi-paradigm. It has curly-bracket syntax, dynamic typing, prototype-based object-orientation, and first-class functions.

- **Bootstrap :**

It is open-source CSS framework directed at responsive, mobile-first front-end web development. It contains CSS and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.

Back End Tools

PHP

In MeroChat, PHP is used for the back-end and for making dynamic web pages. It is used for server side scripting purpose to add connectivity to the database and also used to encrypt the data, validate the user data, confirm user to go to certain pages, login pages. It also includes add, update and delete the data from the database.

Server

APACHE SERVER

In MeroChat, Apache server is used to run php files and creating fast and dynamic web pages.

Database

MYSQL

MySQL is use for storing all the information required to the database in MeroChat. It is used for performing CRUD operation such as create, delete and update data from the database as requested by the user.

Documentation Tools

- **WPS Office**

This is used for writing and editing the documentation of MeroChat.

- **Draw.io**

This is used to generate diagrams for system analysis and design of MeroChat. Diagrams were created using this tool to save time since all components are available with drag and drop functions.

- **Sublime text:**

This is used to write code of MeroChat

4.1.2 Implementation Details of Modules (Description of procedures)

4.2 Testing

4.2.1 Test Cases for Unit Testing

System testing is done by giving different training and testing datasets. This test is done to evaluate whether the system is providing accurate summary or not. During the phase of the development of the system, our system is tested time and again. The series of testing conducted are as follow:

Table 4.2: User Registration testing of MeroChat

Test Case	Scenario	Input	Expected Output	Actual Output	Remarks
1	Register with filling incorrect details	First name: Manoj Last name: 3456789 Email:manoj@gmail.com Password:12345	Registration unsuccesfull.	Only alphabet allowed in last name.	FAIL
2	Register with filling correct	First name: Manoj Last name: Sitoula Email:manoj@gmail.com	Registration successfull.	Registration successfull now you can login.	PASS

	details	Password:12345			
--	---------	----------------	--	--	--

Table 4.3: User Login Testing of MeroChat

Test Case	Scenario	Input	Expected Output	Actual Output	Remarks
1	Signin with filling incorrect details	Email:manoj@gmail.com Password:1234568	Signin unsuccesfull.	Invalid username or password	FAIL
2	Signin with filling correct details	Email:manoj@gmail.com Password:12345	Signin successfull.	Login Successful	PASS

4.2.2 Test Cases for System Testing

- Password field accept minimum 8-16 character, must include at least one numeric value, one uppercase letter, one lowercase letter and one special character.

Table 4.4: Password Acceptence Testing of MeroChat

Test Scenario	Test Scenario description	Expected outcomes
1	Enter less than 8 characters in the password field	System should not accept
2	Enter more than 16 characters in the password field	System should not accept

3	Enter 8-16 characters in the password field with at least 1 uppercase letter, one lowercase letter, 1 digit and 1 special character	System should accept
4	Else	System should not accept

Table 4.5 : Phone Number Acceptance Testing of MeroChat

Test Scenario	Test Scenario description	Expected outcomes
1	Enter less than 10 digits number	System should not accept
2	Enter more than 10 digits number	System should not accept
3	Enter exact 10 digits number	System should accept

Table 4.6: Email Acceptance Testing Of MeroChat

Test Scenario	Test Scenario description	Expected outcomes
1	Address without an @ and a dot	System should not accept
2	Address with an @ but without a dot	System should not accept
3	Address without an @ but with a dot	System should not accept
4	Address with an @ and a dot	System should accept

CHAPTER 5:

CONCLUSION AND FUTURE RECOMMENDATION

5.1 Lesson learnt / outcomes

There have been several improvements in our programming language and writing skills as well as our time management skills while doing this project. It was difficult at the beginning because it was our first project and everything was new. Although it is our first project it turned out to match all the expectations. A lot was learned about proper time management as the project had to be submitted before the deadline along with the documentation. Although it is expectedly good, some new features to this system could be added in the upcoming days to make it more user friendly and efficient.

5.2 Conclusion

We, the project team will design and implement a system capable of allowing users to send and receive message, files and photos like what's app under name 'MeroChat'. By MeroChat, the project team wants users to have easy and secure means of communication with each other, easy platform for sharing files and photos.

5.3 Future recommendations

The Success of this Application depends on the number of the audiences who are willing to use this system. Future developments will be done according to the feedback and reviews obtained from the users of system. Databases will be upgraded and refined frequently. User interfaces will be updated and upgraded according to the needs of the users and changes in technology.

Following are the possible future enhancement:

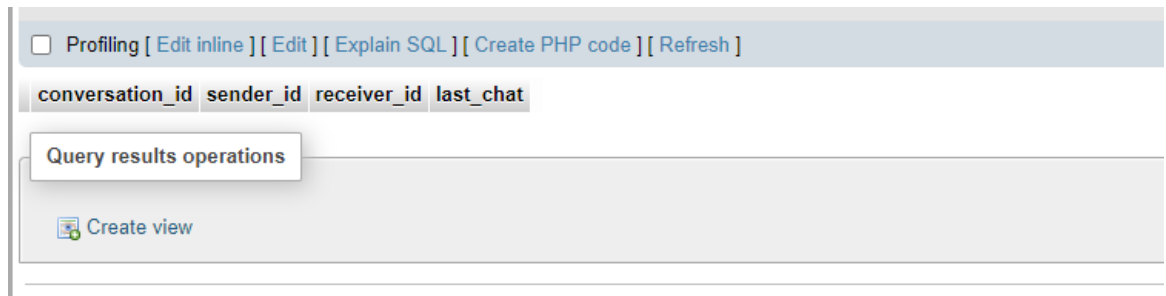
- Audio and video call.
- Voice message.
- Group call.
- Bug Fixes.

APPENDIX:

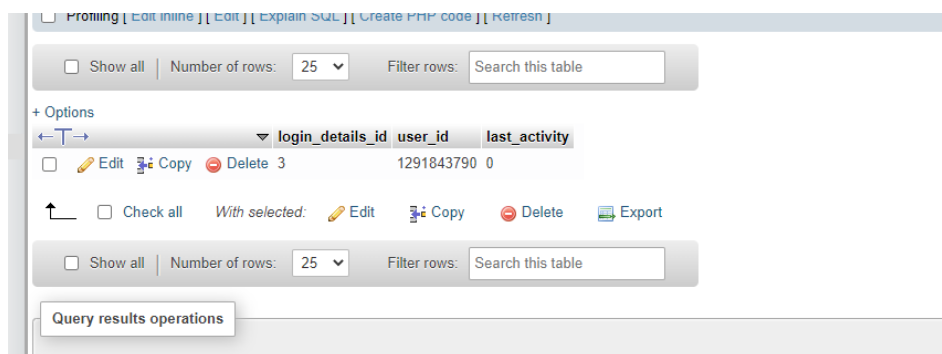
SYSTEM SCREENSHOTS

➤ Database overviews

Conversaation



Login Detail



Message_table



Password_reset

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
SELECT * FROM `password_reset`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

Pid	user_id	token	expDate
-----	---------	-------	---------

Query results operations

 [Create view](#)

Users

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0009 seconds.)

```
SELECT * FROM `users`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

user_id	first_name	last_name	email	password	phone_no	profile_pic	status	verified
---------	------------	-----------	-------	----------	----------	-------------	--------	----------

Query results operations

 [Create view](#)

➤ System Overviews

Login

Mero Chat

Login to your Account.

Email

Password

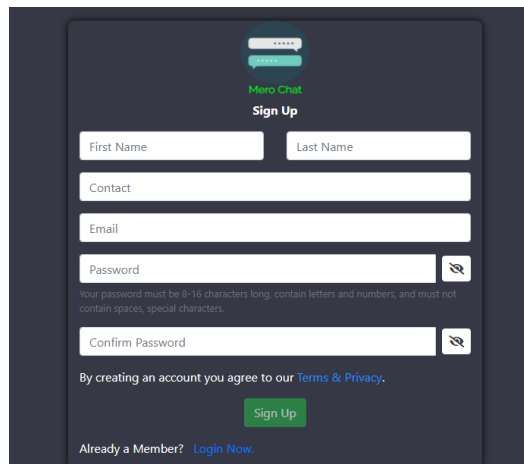
☐ Remember Password.

[Forget Password ?](#)

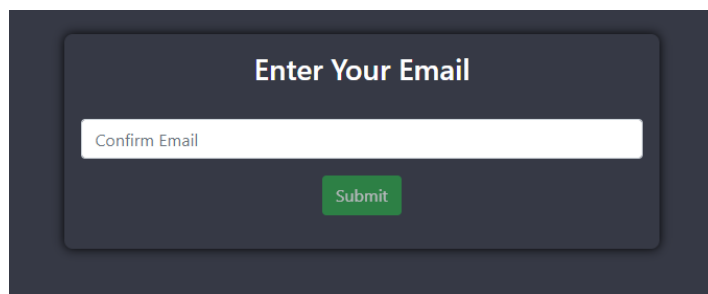
Log In

Not yet a Member? [Signup Now.](#)

Signup

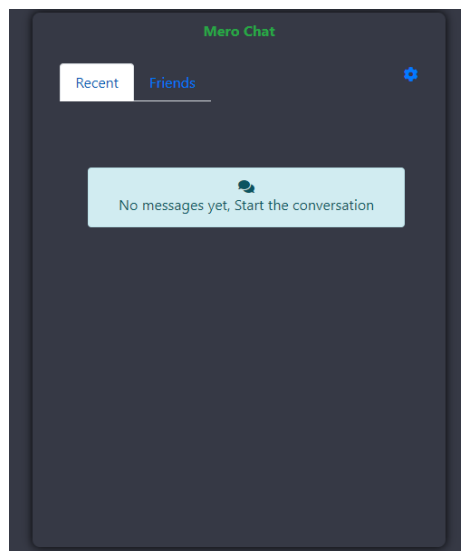


A dark-themed sign-up form for 'Mero Chat'. At the top, there are two tabs: 'Sign Up' (active) and 'Login'. Below the tabs, the 'Mero Chat' logo is displayed. The form contains several input fields: 'First Name', 'Last Name', 'Contact', 'Email', 'Password', and 'Confirm Password'. The 'Password' and 'Confirm Password' fields have a strength indicator icon on the right. A note below the password fields states: 'Your password must be 8-16 characters long, contain letters and numbers, and must not contain spaces, special characters.' Below the form, there is a line of text: 'By creating an account you agree to our [Terms & Privacy](#).' A green 'Sign Up' button is positioned below this text. At the bottom left, there is a link: 'Already a Member? [Login Now](#)'.



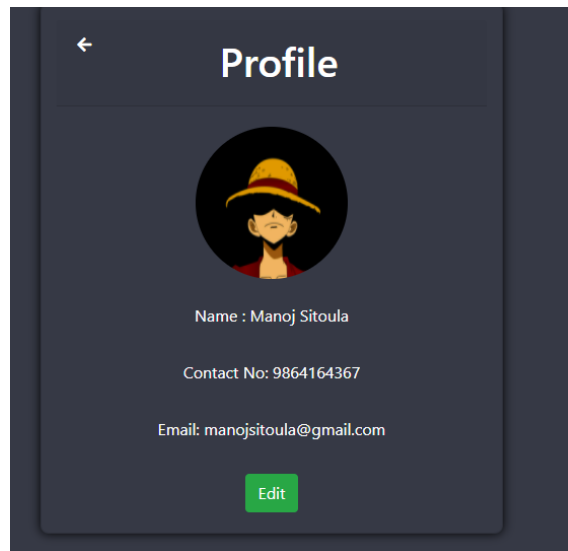
A dark-themed form titled 'Enter Your Email'. It features a single input field labeled 'Confirm Email'. Below the input field is a green 'Submit' button.

Home Page

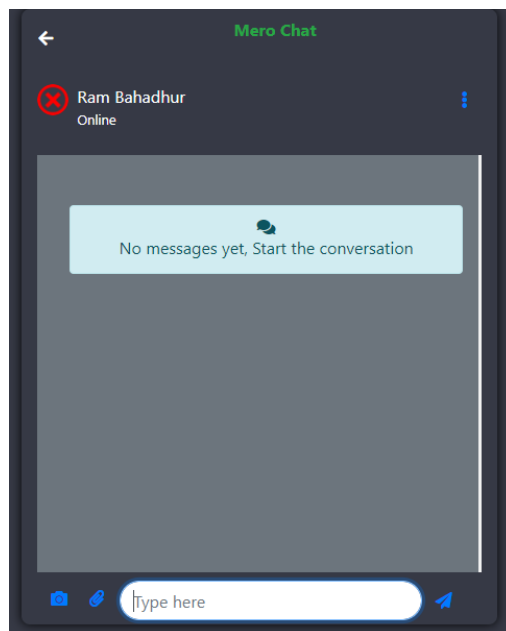


The 'Mero Chat' home page interface. At the top, the 'Mero Chat' logo is centered. Below it, there are two tabs: 'Recent' (active) and 'Friends'. A settings gear icon is located to the right of the tabs. The main content area is a large, light blue rectangle with a speech bubble icon and the text: 'No messages yet, Start the conversation'.

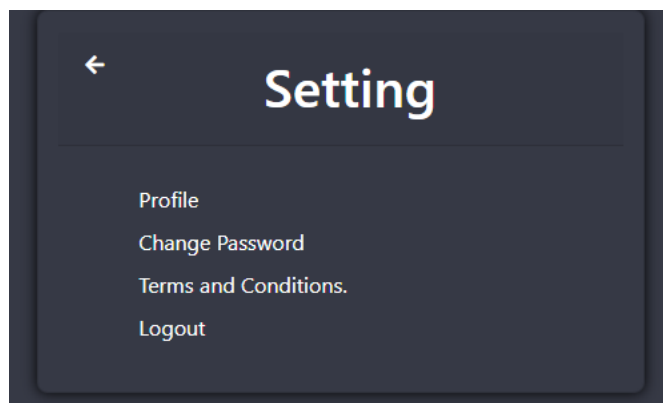
Profile page



Chat Page



Setting





Change Password

Old Password



New Password



Your password must be 8-16 characters long, contain letters and numbers, and must not contain spaces, special characters.

Confirm New Password



Change Password

REFERENCES

1. Siegler, MG(15 November 2010) "Facebook's Moders Messaging System: Seamless, History, And A Social Inbox".TechCrunch,AOL.
2. "WhatsApp Blog". WhatsApp.com.
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